

Navigating Market Shifts: PEL's Dilemma

On January 1, 2007, PowerFlow Engineering Ltd (PEL) held a key meeting with its top decision-makers to discuss the company's future direction. The meeting included the MD, Ashok Rathod and the Senior VP of Sales and Marketing, Mr. Arun Mathew. By this time, PEL had already established a strong reputation in the industrial sector, particularly with its screw and reciprocating compressors. However, the leadership team recognized that the market was evolving, with industries increasingly turning to centrifugal compressors for their higher flow rates and energy efficiency. While PEL had been importing bare centrifugal compressors (annexure-2), it became clear that to stay competitive, the company needed to develop its own in-house bare centrifugal compressors. Ashok Rathod saw this shift as a golden opportunity to expand PEL's product portfolio and capture a larger share of the market. However, Mathew raised concerns about the complexities of entering the centrifugal compressor market, including the technological challenges, the need for substantial investment in manufacturing capabilities, and the uncertainty of the company's ability to effectively market the new product. Additionally, he questioned how PEL could build a strong reputation for a product it had not previously manufactured and how it would differentiate itself in an increasingly competitive market. With these mixed opinions, the team found themselves facing a critical dilemma—whether to proceed with building an in-house bare centrifugal compressor line to capture the market or to stick with their existing products and strengthen their position there.

About PEL LTD

PEL was founded in 1978 and started manufacturing of air compressors. Their offerings include: Screw compressor and Reciprocating Compressor with capacities ranging from 30 to 10,000 Cubic Feet Per Minute (CFM).

New product lines were then added, including Air Conditioning and Refrigeration systems, Marine HVACR, Process Gas systems, Vapour Absorption System and Hydraulic Power Transmission machinery. PEL is widely spread into the domestic market and is trying to cover the international market as well. This company earned an enviable reputation for its systems engineering. Over the years, PEL has developed various sophisticated and high-tech products in the above categories to cater to the demands of various industrial sectors; domestic, as well as international. This company has experience in developing a range of advanced products tailored to Indian conditions, consistently upgrading them to uphold the highest standards of quality and reliability. The joint venture details are as follows:

As of 2007, PEL had established a notable global presence, particularly through its joint ventures and international collaborations:

- PEL had a joint venture with Demag Group, which was later acquired by Siemens AG, Germany.
- In 2003, PEL Ltd sold its stake in Demag Pvt. Ltd. to Siemens AG.
- PEL had a joint venture with McQuay International, USA. In 2006-07, this joint venture agreement was terminated, and the company was renamed as PEL Chillers Private Limited.

These collaborations indicate PEL's active engagement in international markets during **that** period. Additionally, it formed several joint ventures and technology partnerships with renowned global companies. PEL offers various services namely, Installation Support, Maintenance Services, Operation and Maintenance (O&M) Contracts and Logistic Services followed by air audits.

PEL Vision

To lead the industrial compressor market by delivering innovative, energy-efficient, competitive and reliable solutions that empower industries to achieve operational excellence, while fostering sustainable growth through cutting-edge technology and a commitment to quality.

About the Product - Bare Centrifugal Compressor

A bare centrifugal compressor refers to a centrifugal compressor without additional components such as drivers (motors or turbines), lubrication systems, control panels, or auxiliary equipment. Bare centrifugal compressors are typically integrated into engineered systems where they are paired with external components to suit specific operational needs. For instance, in an oil refinery, a bare centrifugal compressor might be coupled with a steam turbine driver and a custom control system to handle high-pressure gas compression. Bare centrifugal compressors were imported during 2007 as indigenous development was minimal. They also used in air and gas compressor applications due to their high efficiency and low maintenance.

Govt of India Initiatives in 2007 for in-house development:

There were numerous initiatives announced by Govt of India in the year 2007. Some of the them were:

- Promoting domestic manufacturing in the year 2007.
- Announced Incentives for research and development, such as weighted tax deductions (up to 150%) under Section 35(2AB) of the Income Tax Act, were available to companies involved in in-house R&D.
- Special Economic Zones (SEZs)
- Tax incentives (e.g., 100% income tax exemption for the first five years).
- Simplified procedures for business operations

Company Financials

The PEL Compound Annual Growth Rate (CAGR) of revenue from 2001 to 2006 is as follows:

- a) Start Revenue (2001): ₹187.5 Crores
- b) End Revenue (2006): ₹260.0 Crores
- c) Number of Years: 5
- d) PEL grew 6.76% CAGR during this period.

Profit and Loss 2001 to 2006(overall business)

Fiscal Year	Revenue (₹ Crores)	Profit Before Tax (₹ Crores)	Profit After Tax (₹ Crores)
2000-2001	187.5	16.2	10.5
2001-2002	195.0	18.0	11.7
2002-2003	210.0	20.5	13.3
2003-2004	225.0	22.8	14.8
2004-2005	240.0	25.2	16.4
2005-2006	260.0	28.0	18.2

Industry Outlook/Competitors

In 2007, PEL was a prominent player in India's industrial machinery sector, particularly in the compressor market. Its main competitors included:

Ingersoll Rand (India) Ltd.: A subsidiary of the global diversified industrial firm, offering a wide range of industrial solutions, including compressors.

Atals Copco

Atlas Copco was indeed a significant competitor to PEL in the Indian market by 2007. It was a global leader in compressors and industrial solutions, had a strong presence in India, offering products such as: Air Compressors (Reciprocating and Screw Compressors), Gas Compressors and Industrial Tools and Equipment.

ELGI Equipments Ltd: A leading manufacturer of air compressors and related products in India.

Market share of PEL and competitors by 2007.

Company	Market Share (India, 2007)	Product Focus	Key Strengths
PEL	Miniscule	Air Compressors, Refrigeration Compressors	Strong presence in India, Established brand, Focused on refrigeration and industrial compressors
Atlas Copco	Approx. 30-40% in India (2007 estimate)	Air Compressors, Industrial Equipment, Tools	Global presence, Innovation in industrial solutions, Strong foothold in India since 2007



Ingersoll Rand (India) Ltd.	Approx. 50-60% in India (2007 estimate)	Air Compressors, Vacuum Pumps	Diverse product range, Focus on quality and innovation, Leading in India's compressor market
ELGI Equipments Ltd	Approx. – 30%	Air compressors and allied equipment.	Air compressors, with an emphasis on providing energy-efficient, durable, and reliable solutions for industrial and commercial applications

Suppliers

Hamilton International Corporation: Hamilton was a prominent provider of flow equipment products, systems, and services to the global oil, gas, and process industries along with the Bare Centrifugal compressors.

Limitations or Challenges

a) Risk of technology transfer

Technology is one of the challenges in the area of bare centrifugal compressors. These includes; Insufficient access to high-precision CNC machines, 5-axis machining, and specialized materials (such as stainless steel and titanium).

b) Lack of expertise

The advanced design and manufacturing of centrifugal compressors required specialized knowledge in aerodynamics, materials science, and precision engineering. The PEL had limited experience in this area. In-house building requires specialized skilled related to the design, operation, and maintenance of advanced compressor technologies.

For initiating the setup, the following manpower estimated:

Design – 10 No's

Operations – 14 No's

Maintenance – 2 No's

c) Investment

The high cost of importing advanced manufacturing equipment (such as CNC machines and 5-axis machining) and establishing state-of-the-art production facilities posed a significant financial risk.

estimated investment cost – (approximate)– Rs. 35.0 Cr

d) Supply Chain Risk:

Building a new network is one of the challenging task. Building a supply chain network across the country incurs lot of time.

Approximate time required for establishing the network across India was 18 months.

e) Timelines and schedule

Due to the lack of expertise and technology, the on-time production and delivery was challenging and might incur penalties if any order accepted without strong estimation.

Questions

1. Should the company go ahead with New Product Development (NPD)? If yes, give a strategy to get market share of 20%? Additionally, estimate the timeframe required to attain this market share, considering key milestones and market dynamics.
2. How new technology can act as a catalyst in NPD?
3. What are the potential risks PEL would face? Especially the situations like covid, demonetization, change in tax structure etc.
4. Establishing the supply chain network was projected to take approximately 18 months. How can it be this achieved?
5. How does developing in-house centrifugal compressors align with PEL's vision?

Annexure -1: Technologies in centrifugal compressors and its associated costs.

1. **Impeller Design:** The impeller is a critical component that imparts velocity to the gas. Advancements in impeller design during this period focused on enhancing aerodynamic efficiency to improve overall compressor performance (F1000Research)
2. **Diffuser Optimization:** The diffuser converts the kinetic energy from the impeller into pressure. Research in 2007 explored the effects of impeller-diffuser interactions on compressor performance, aiming to optimize this component for better efficiency (ResearchGate)
3. **Integrally Geared Compressors:** These compressors feature multiple stages mounted on a single gearbox, allowing for higher efficiency and compact design. They were increasingly adopted in various industries for their operational advantages (Transparency market research)

Annexure 2: Bare Centrifugal Compressor Cost

Bare compressor Import cost per item @ Rs. 45L ~ 60 L (Inclusive taxes and duties)

Imports from Hamilton International Corporation (2001 to 2006)

Year	Count	Profit (%)
2001	20	15
2002	30	15.5
2003	35	16.25
2004	40	16.3
2005	44	16.3
2006	48	16.6

Annexures3 Glossary

Compressor

A mechanical device used to increase the pressure of a gas by reducing its volume.

Centrifugal Compressor

A type of dynamic compressor that uses a rotating impeller to add velocity to the gas and then converts this velocity into pressure.

Dynamic Compressor

Compressors that impart kinetic energy to the gas, which is then converted to pressure energy, such as centrifugal and axial compressors.

A **bare centrifugal compressor** refers to a centrifugal compressor unit that is provided without auxiliary systems, casing, or additional components like drivers, controls, or lubrication systems. Essentially, it is the core mechanical component of the compressor, typically including the **impeller**, **diffuser**, and other internal parts essential for the compression process.

Reciprocating Compressor

A positive displacement compressor that uses a piston within a cylinder to compress the gas.

Key Features of Bare Centrifugal Compressors:

1. Core Component Only

It includes just the basic compression mechanism, often used as a part of a larger system.

2. Custom Integration

These compressors are often purchased by Original Equipment Manufacturers (OEMs) or system integrators who add their own casings, motors, controls, and auxiliary equipment.

3. Flexible Application

Bare centrifugal compressors can be tailored and integrated into various applications, such as refrigeration systems, gas pipelines, or industrial processes.

4. Cost-Effectiveness

By excluding auxiliary components, these compressors may reduce initial costs for customers planning to use their existing infrastructure or components.

5. Specialized Use

They are often employed in scenarios requiring custom setups or specific performance characteristics.

Screw compressor

A **screw compressor** is a type of **rotary positive displacement compressor** that uses two interlocking helical screws (called rotors) to compress gas or air. As the screws rotate, the volume of trapped gas decreases, leading to an increase in pressure.

Reciprocating compressor

A **reciprocating compressor** is a type of **positive displacement compressor** that uses a piston moving back and forth (reciprocating) inside a cylinder to compress gas or air.